METABOLISM OF AMINO ACIDS

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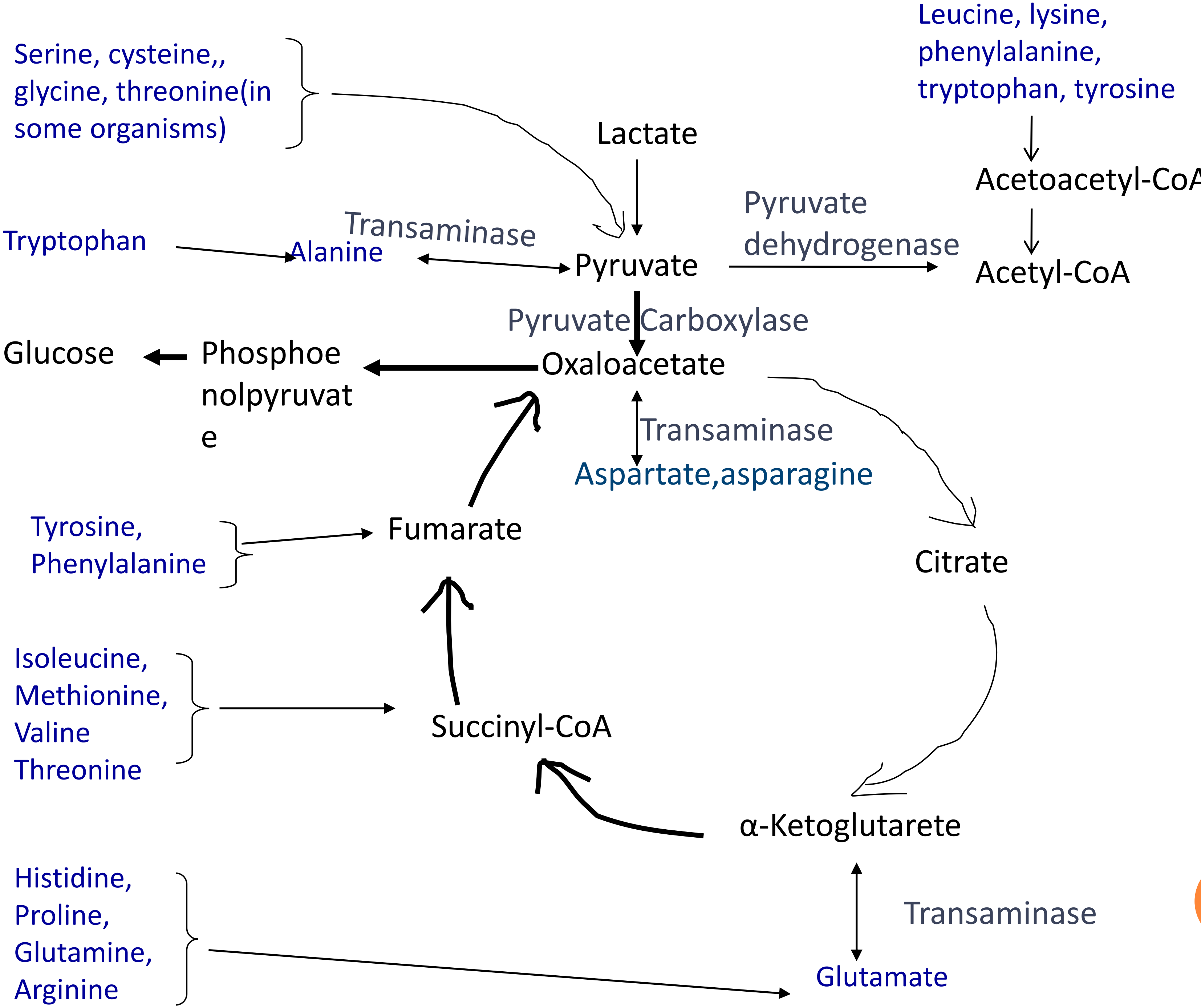
CATABOLISM OF THE CARBON SKELETON OF AMINO ACIDS

- The pathways of amino acid catabolism normally account for only 10-15% of the human bodies energy requirements
- The 20 catabolic pathways converge to form only five products all of which enter the citric acid cycle
- From here the carbon skeletons are diverted to gluconeogenesis or ketogenesis or are completely oxidized to CO_2 and H_2O


Serine, cysteine,, glycine, threonine(in some organisms)
Leucine, lysine, phenylalanine, tryptophan, tyrosine
Lactate
Acetoacetyl-CoA
Pyruvate dehydrogenase
Tryptophan
Transaminase
Alanine
Pyruvate
Acetyl-CoA
Pyruvate Carboxylase
Glucose
Phosphoe nolpyruvat e
Oxaloacetate
Transaminase
Aspartate,asparagine
Tyrosine, Phenylalanine
Fumarate
Citrate
Isoleucine, Methionine, Valine Threonine
Succinyl-CoA
α-Ketoglutarete
Histidine, Proline, Glutamine, Arginine
Transaminase
Glutamate

- All or part of the carbon skeletons of ten amino acids are ultimately broken down to acetyl-CoA which enters the citric acid cycle
  - Five are converted to acetyl-CoA and/or acetoacetyl-CoA, which is then cleaved to acetyl-CoA;

    Leucine ,lysine,phenylalanine, tryptophan, tyrosine
  - **Five amino acids enter via pyruvate;**

    Alanine, tryptophan, cysteine, serine, glycine
- Five are converted to α-ketoglutarate:

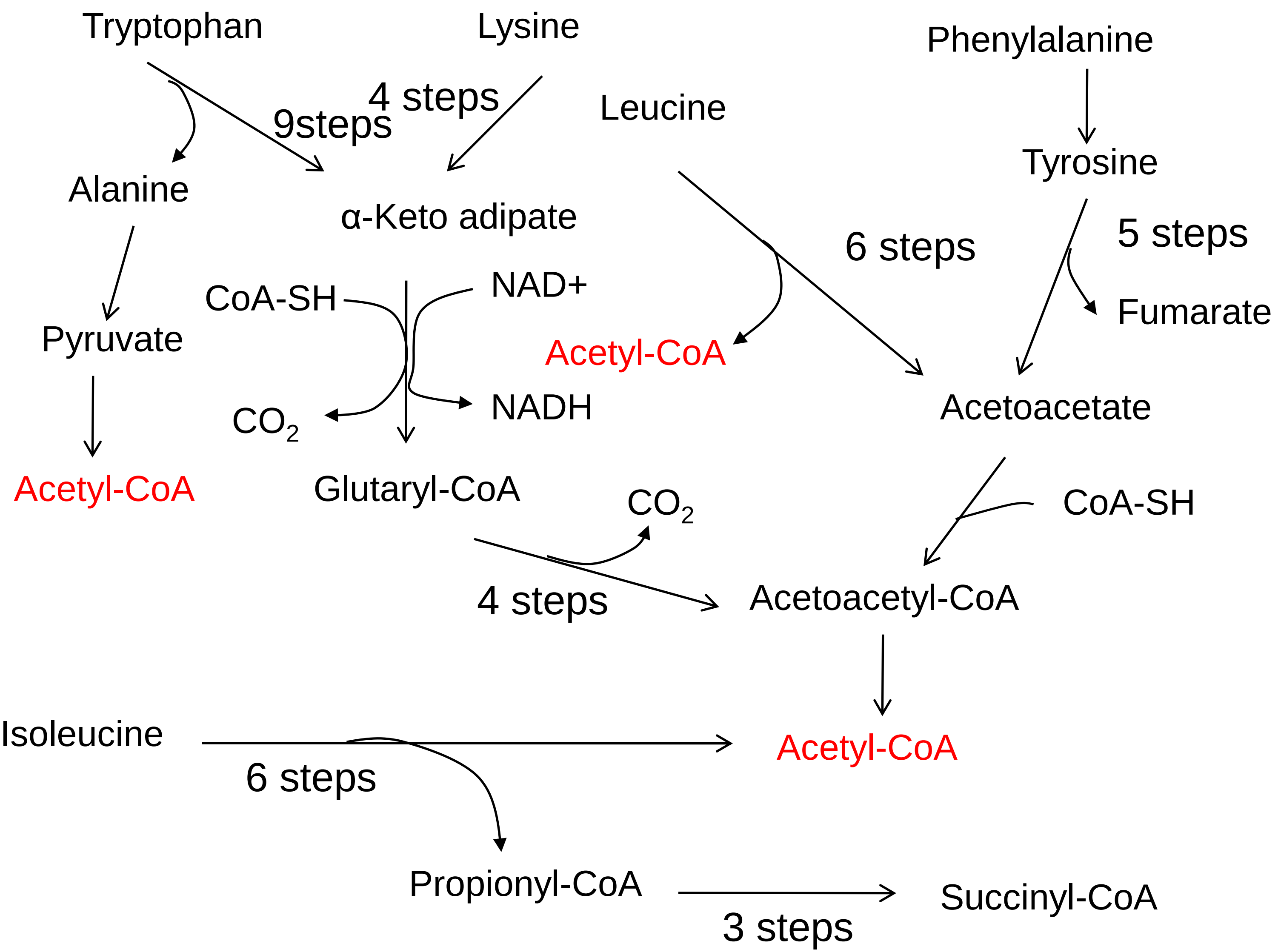
  Arginine, glutamine , glutamate, histidine, proline
- Four are converted to succinyl-CoA

  Isoleucine, methionine, threonine, valine
- Two are converted to fumarate

  Phenylalanine, tyrosine
- Two are converted to oxaloacetate

  Asparagine, aspartate

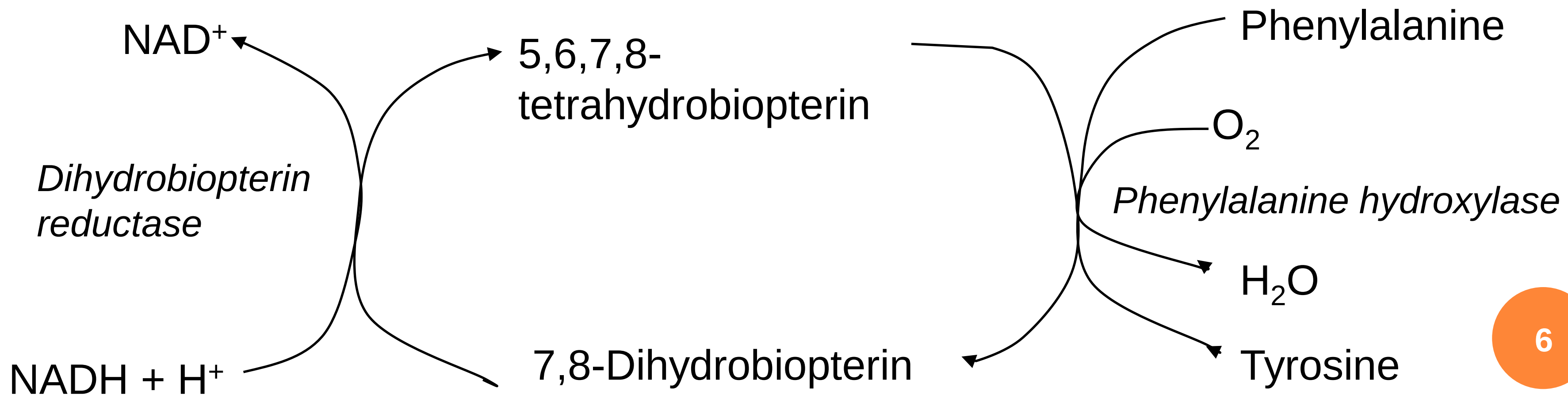
2. Catabolic pathways for tryptophan, lysine, phenylalanine, tyrosine, leucine, and isoleucine



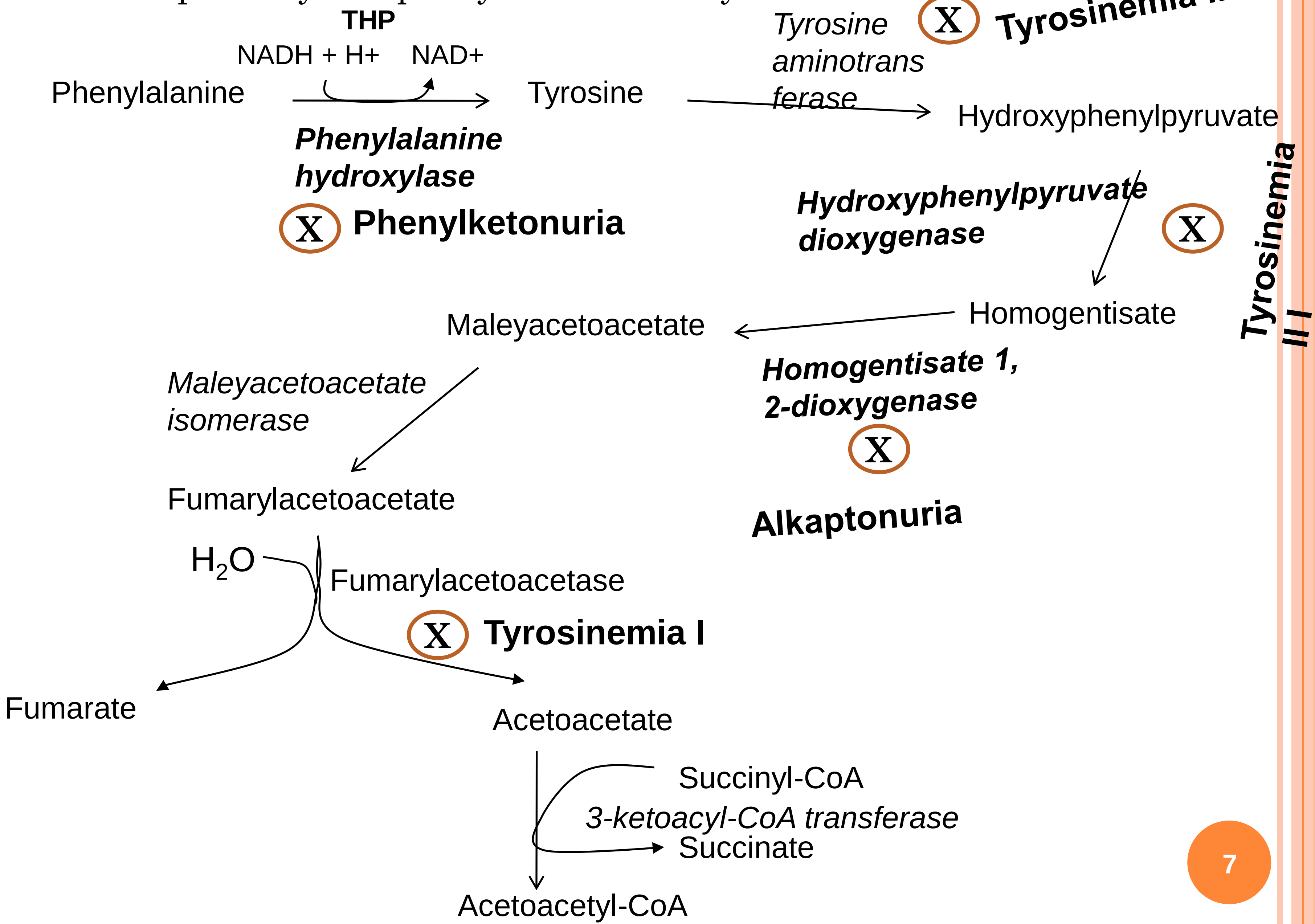
- Tryptophan yields acetyl-CoA via both pyruvate and acetoacetyl-CoA
  - Some intermediates of tryptophan catabolism are precursors of biomolecules such as nicotinic acid (a precursor of NAD and NADP in animals) and seroton (a neurotransmitter in vertebrates)

Phenylalanine catabolism is genetically defective in some people

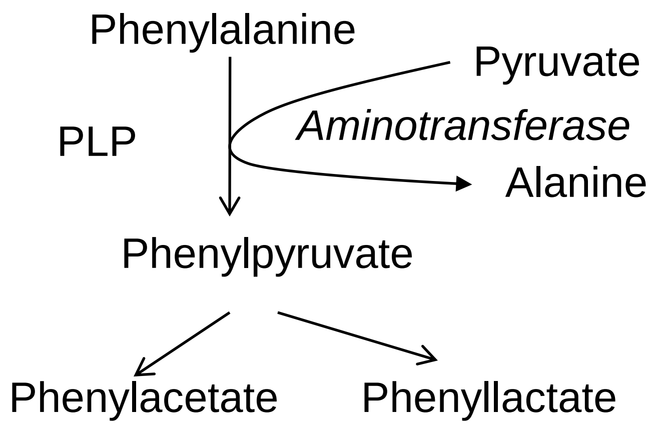
- A genetic defect in phenylalanine hydroxylase the first enzyme in the catabolic pathway of phenylalanine is responsible for the disease phenylketonuria (PKU)
- Phenylalanine hydroxylase
  - is a mixed –function oxidase
  - Requires the cofactor **tetrahydrobiopterin** which carries electrons from NADH to O_2 and becomes oxidized to dihydrobiopterin
  - The cofactor is reduced by dihydrobiopterin reductase



Catabolic pathways for phenylalanine and tyrosine



- In individuals with PKU, a secondary pathway of phenylalanine metabolism comes into play in which phenylalanine undergoes transamination with pyruvate to yield phenylpyruvate
  - Phenylalanine and phenylpyruvate accumulate in the blood and tissues and are excreted in the urine
  - Much of the phenylpyruvate, rather than being excreted as such, is either decarboxylated to phenylacetate or reduced to phenyllactate
  - **Phenylacetate gives the urine a characteristic odor**



- Accumulation of phenylalanine or its metabolites leads to mental retardation
- Phenylketonuria can also be caused by a defect in the enzyme that catalyzes the regeneration of tetrahydrobiopterin

In alkaptonuria the defective enzyme is homogentisate dioxygenase

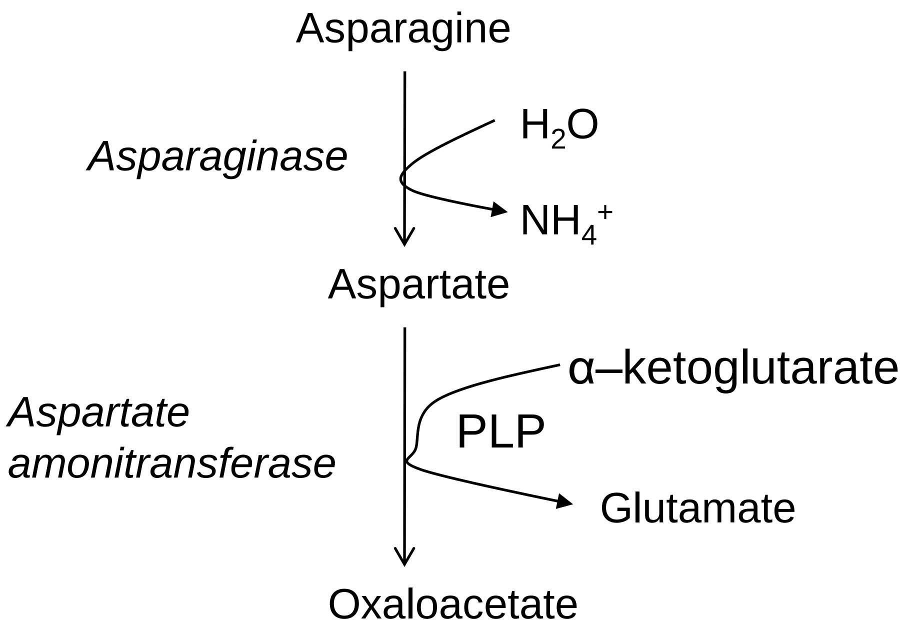
- **Less serious than PKU**
- **Large quantities of homogentisate are excreted and its oxidation turns the urine black**

Branched chain amino acids are not degraded in the liver

- Although much of the catabolism of amino acids takes place in the liver, the three amino acids with branched side chains (**leucine, isoleucine and valine)** are oxidized as fuels primarily in muscle, adipose tissue, kidney, and brain tissue
- The extrahepatic tissues contain an aminotransferase absent in the liver, that acts on all three branched-chain amino acids to produce corresponding α-ketoacids
- The ***branched-chain α-keto acid dehydrogenase complex*** then catalyzes the oxidative decarboxylation of all three α-ketoacids, in each case releasing a carboxyl group as CO_2 and producing the acyl-CoA derivative (Analogous to reactions catalyzed by pyruvate dehydrogenase and α–ketoglutarate dehydrogenase)
  - The three enzyme systems have the same cofactors (NAD, FAD, TPP, lipoate, CoA)
  - The branched-chain α-keto acid dehydrogenase complex is regulated by covalent modification
    - Phosphorylation deactivates the enzyme
    - Dephosphorylation activates it
- In **maple syrup urine disease** (because of the characteristic ordor imparted to the urine by the α-ketoacids) the three branched-chain α-ketoacids accumulate in the blood and spill over in the urine
  - Results from defective ***branched-chain α-keto acid dehydrogenase complex***
  - Untreated this condition leads to abnormal development of the brain, mental retardation, and death in early infancy
  - Treatment involves limiting the amounts of valine leucine and isoleucine to the minimum required for normal growth

Asparagine and aspartate are degraded to oxaloacetate

- The carbon skeletons of asparagine and aspartate ultimately enter the citric acid cycle as oxaloacetate
- Asparaginase catalyzes the hydrolysis of asparagine to aspartate, which undergoes transamination with α–ketoglutarate to yield glutamate and oxaloacetate



Some amino acids can be converted to ketone bodies and others to glucose

- The six amino acids degraded to acetoacetyl-CoA and /or acetyl-CoA−**tryptophan, phenylalanine, tyrosine, isoleucine, leucine and lysine** can yield ketone bodies in the liver, where acetoacetyl-CoA is converted to acetoacetate and β-hydroxybutyrate. Ketogenic amino acids
- Glucogenic amino acids can be converted to pyruvate, α-ketoglutarate, succinyl-CoA, fumarate, and /or oxaloacetate
- **Tryptophan, phenylalanine, tyrosine, isoleucine,** are both ketogenic and glucogenic